

Snowflake + GitHub Actions Authentication — Developer Cheatsheet

Connect GitHub Actions to Snowflake securely: OIDC (secretless, recommended) or Key Pair/PAT as fallback for older tooling.

[!IMPORTANT] Community cheatsheet — not official Snowflake documentation. For the authoritative reference, see [Snowflake CLI GitHub Action docs](#).

Table of Contents

- [Choose Your Method](#)
- [OIDC Setup: Snowflake Side](#)
- [OIDC Setup: GitHub Actions](#)
- [OIDC Subject Patterns](#)
- [Network Policy](#)
- [Key Pair Auth](#)
- [PAT / Password \(Legacy\)](#)
- [Env Vars Reference](#)
- [Cleanup](#)
- [Tips & Gotchas](#)
- [References](#)

Choose Your Method

Method	GitHub Secrets needed	Credential rotation	Snow CLI
OIDC (recommended)	SNOWFLAKE_ACCOUNT only	None — tokens short-lived per run	≥ 3.11
Key pair	Account + user + private key	Yes — rotate private key	Any
PAT / Password	Account + user + password/PAT	Yes — manual rotation	Any

Check your installed version:

```
# run in terminal
snow --version # must be ≥ 3.11 for OIDC; upgrade: pip install -
U snowflake-cli
```

[!TIP] OIDC is the only method that stores zero long-lived secrets in GitHub. Use it whenever Snow CLI ≥ 3.11 is available.

OIDC Setup: Snowflake Side

Run as ACCOUNTADMIN. Creates a service user that trusts GitHub's OIDC tokens directly — no password, no key pair stored anywhere.

USE ROLE ACCOUNTADMIN;

```
-- 1. Least-privilege role for the workflow
CREATE ROLE IF NOT EXISTS gh_actions_role
  COMMENT = 'Least-privilege role for GitHub Actions via OIDC';
```

```
GRANT USAGE ON WAREHOUSE my_wh TO ROLE gh_actions_role;
-- Add further grants only as required by the workflow
```

```
-- 2. Service user mapped to GitHub OIDC
CREATE USER IF NOT EXISTS gh_actions_svc_user
  TYPE = SERVICE
  WORKLOAD_IDENTITY = (
    TYPE = OIDC
    ISSUER = 'https://token.actions.githubusercontent.com'
    SUBJECT = 'repo:<owner>/<repo>;environment:<env-name>'
  )
  DEFAULT_ROLE = gh_actions_role
  DEFAULT_WAREHOUSE = my_wh
  COMMENT = 'GitHub Actions OIDC service user';
```

```
GRANT ROLE gh_actions_role TO USER gh_actions_svc_user;
```

Verify the user and workload identity were created correctly:

```
DESCRIBE USER gh_actions_svc_user;
SHOW PARAMETERS LIKE 'DEFAULT_ROLE' FOR USER gh_actions_svc_user;
```

Replace <owner>/<repo>;environment:<env-name> with your values — see [OIDC Subject Patterns](#).

OIDC Setup: GitHub Actions

Two requirements: permissions: id-token: write on the job and use-oidc: true on the action. Only one GitHub Secret is needed: SNOWFLAKE_ACCOUNT.

```
name: Deploy to Snowflake
on:
  push:
    branches: [main]

permissions:
  id-token: write    # required for OIDC token issuance
  contents: read

jobs:
  deploy:
    runs-on: ubuntu-latest
    environment: prod    # must match the SUBJECT in CREATE USER
    steps:
      - uses: actions/checkout@v4
        with:
          persist-credentials: false

      - uses: snowflakedb/snowflake-cli-action@v2
        with:
          use-oidc: true
          cli-version: "3.16"

      - name: Test connection
        env:
          SNOWFLAKE_ACCOUNT: ${ secrets.SNOWFLAKE_ACCOUNT }
        run: snow connection test -x

      - name: Run governance SQL
        env:
          SNOWFLAKE_ACCOUNT: ${ secrets.SNOWFLAKE_ACCOUNT }
        run: snow sql -x -f governance/policy.sql
```

-x (--temporary-connection) builds the connection from env vars — no config.toml required on the runner.

Set the one required GitHub Secret:

```
# run in terminal
gh secret set SNOWFLAKE_ACCOUNT --body "orgname-accountname"
```

OIDC Subject Patterns

The SUBJECT in CREATE USER must match the claim GitHub emits for the job exactly.

Subject format	Matches	Job requirement
repo:<owner>/ <repo>:environment:<name>	Job targets a named environment	environment: <name> set on the job (recommended)
repo:<owner>/ <repo>:ref:refs/heads/ <branch>	Push to the specified branch	on: push, no environment: on job
repo:<owner>/ <repo>:pull_request	Any pull request event	on: pull_request, no environment:

[!NOTE] When a job sets environment:, GitHub always uses the environment-based subject regardless of trigger. Use environment: subjects for tighter scoping — one OIDC user per environment (e.g. dev, prod).

Network Policy

Only needed if your Snowflake account restricts inbound IPs. Use the Snowflake-managed rule — no manual CIDR list maintenance required.

Check whether the service user already has a network policy:

```
SHOW PARAMETERS LIKE 'NETWORK_POLICY' FOR USER  
gh_actions_svc_user;
```

Create and apply a new policy with the managed GitHub Actions rule:

```
CREATE NETWORK POLICY gh_actions_policy  
ALLOWED_NETWORK_RULE_LIST =  
( 'SNOWFLAKE.NETWORK_SECURITY.GITHUBACTIONS_GLOBAL' );
```

```
ALTER USER gh_actions_svc_user  
SET NETWORK_POLICY = gh_actions_policy;
```

Add to an existing policy instead of replacing it:

```
ALTER NETWORK POLICY existing_policy  
ADD ALLOWED_NETWORK_RULE_LIST =  
( 'SNOWFLAKE.NETWORK_SECURITY.GITHUBACTIONS_GLOBAL' );
```

[!NOTE] If you see Incoming request with IP is not allowed to access Snowflake — add the network policy. If the account has no IP restrictions, skip this section.

Key Pair Auth

Use when Snow CLI < 3.11 or OIDC is not available. GitHub Secrets needed: SNOWFLAKE_ACCOUNT, SNOWFLAKE_USER, SNOWFLAKE_PRIVATE_KEY_RAW.

Generate a key pair:

```
# run in terminal
openssl genrsa 2048 | openssl pkcs8 -topk8 -nocrypt -out
private_key.p8
openssl rsa -in private_key.p8 -pubout -out public_key.pub
```

Register the public key on the Snowflake user:

```
ALTER USER my_user SET RSA_PUBLIC_KEY =
    '<public_key.pub contents, no header/footer>';
```

Store all three required GitHub Secrets:

```
# run in terminal
gh secret set SNOWFLAKE_ACCOUNT      --body "orgname-
accountname"
gh secret set SNOWFLAKE_USER        --body "my_user"
gh secret set SNOWFLAKE_PRIVATE_KEY_RAW < private_key.p8
```

Temporary connection (no config.toml — recommended for CI):

```
- uses: snowflakedb/snowflake-cli-action@v2
  with:
    cli-version: "3.16"

- name: Deploy
  env:
    SNOWFLAKE_AUTHENTICATOR: SNOWFLAKE_JWT
    SNOWFLAKE_ACCOUNT:       ${ secrets.SNOWFLAKE_ACCOUNT }
    SNOWFLAKE_USER:          ${ secrets.SNOWFLAKE_USER }
    SNOWFLAKE_PRIVATE_KEY_RAW: $
                             {{ secrets.SNOWFLAKE_PRIVATE_KEY_RAW }}
  run: snow connection test -x
```

Named connection (commit a minimal config.toml, override via env vars):

```
# config.toml – commit this, contains no secrets
default_connection_name = "ci"
```

[connections.ci]

- uses: snowflakedb/snowflake-cli-action@v2
with:
cli-version: "3.16"
default-config-file-path: "config.toml"
- name: Deploy
env:
SNOWFLAKE_CONNECTIONS_CI_AUTHENTICATOR: SNOWFLAKE_JWT
SNOWFLAKE_CONNECTIONS_CI_ACCOUNT: \$
{{ secrets.SNOWFLAKE_ACCOUNT }}
SNOWFLAKE_CONNECTIONS_CI_USER: \$
{{ secrets.SNOWFLAKE_USER }}
SNOWFLAKE_CONNECTIONS_CI_PRIVATE_KEY_RAW: \$
{{ secrets.SNOWFLAKE_PRIVATE_KEY_RAW }}
run: snow connection test

PAT / Password (Legacy)

Use only for legacy tooling that cannot use OIDC or key pairs.
GitHub Secrets needed: SNOWFLAKE_ACCOUNT, SNOWFLAKE_USER,
SNOWFLAKE_PASSWORD (or PAT).

Generate a PAT in Snowsight (Profile → Programmatic Access
Tokens → Generate) or via SQL:

```
ALTER USER my_user ADD PROGRAMMATIC ACCESS TOKEN my_ci_token  
ROLE_RESTRICTION = my_ci_role;
```

GitHub Actions workflow:

- uses: snowflakedb/snowflake-cli-action@v2
with:
cli-version: "3.16"
- name: Run SQL
env:
SNOWFLAKE_ACCOUNT: \${{ secrets.SNOWFLAKE_ACCOUNT }}
SNOWFLAKE_USER: \${{ secrets.SNOWFLAKE_USER }}
SNOWFLAKE_PASSWORD: \${{ secrets.SNOWFLAKE_PASSWORD }}
run: snow connection test -x

Remove a PAT when no longer needed:

```
ALTER USER my_user REMOVE PROGRAMMATIC ACCESS TOKEN my_ci_token;
```

Env Vars Reference

Variables set automatically by snowflake-cli-action@v2 when use-oidc: true. You only need to provide SNOWFLAKE_ACCOUNT — everything else is injected by the action.

Variable	Value set by action
SNOWFLAKE_AUTHENTICATOR	WORKLOAD_IDENTITY
SNOWFLAKE_WORKLOAD_IDENTITY_PROVIDER	OIDC
SNOWFLAKE_AUDIENCE	snowflakecomputing.com
SNOWFLAKE_TOKEN	GitHub OIDC token (short-lived, per run)

Override the token variable for named connections:

```
- uses: snowflakedb/snowflake-cli-action@v2
  with:
    use-oidc: true
    oidc-token-name: SNOWFLAKE_CONNECTIONS_PROD_TOKEN
```

Use oidc-token-name: SNOWFLAKE_CONNECTIONS_<NAME>_TOKEN where <NAME> is the uppercased connection name from config.toml (e.g. [connections.prod] → PROD).

Cleanup

Remove OIDC objects when decommissioning a workflow:

```
USE ROLE ACCOUNTADMIN;

DROP USER IF EXISTS gh_actions_svc_user;
DROP ROLE IF EXISTS gh_actions_role;
DROP NETWORK POLICY IF EXISTS gh_actions_policy;
```

Remove GitHub Secrets via CLI:

```
# run in terminal
gh secret delete SNOWFLAKE_ACCOUNT
gh secret delete SNOWFLAKE_USER # key pair / PAT only
gh secret delete SNOWFLAKE_PRIVATE_KEY_RAW # key pair only
gh secret delete SNOWFLAKE_PASSWORD # PAT / password only
```

Tips & Gotchas

- **Subject must match exactly.** A mismatch between SUBJECT in CREATE USER and the claim GitHub emits silently fails auth with

a generic error. Test with `snow connection test -x` in a throwaway workflow run before building the full pipeline.

- **environment: overrides all other subject formats.** Once a job sets `environment:`, GitHub always uses the environment claim regardless of trigger (push, PR, schedule). Plan one OIDC service user per environment (dev, staging, prod), not per branch.
- **-x is required in CI.** Without `--temporary-connection`, Snow CLI looks for `~/.snowflake/config.toml` which does not exist on a fresh GitHub Actions runner.
- **Network policy is evaluated before OIDC token validation.** An IP-blocked runner shows `IP not allowed` — not an auth error. Add `SNOWFLAKE.NETWORK_SECURITY.GITHUBACTIONS_GLOBAL` first, then debug the OIDC subject.
- **PATs expire silently.** A workflow using an expired PAT fails without a clear error until someone investigates. Set a calendar reminder before `DEFAULT_EXPIRY_IN_DAYS`. OIDC tokens are always fresh — one more reason to prefer OIDC.

References

- [Snowflake CLI GitHub Action](#)
- [Integrating CI/CD with Snowflake CLI](#)
- [snow auth oidc commands](#)
- [Snowflake Programmatic Access Tokens](#)
- [Snowflake Network Rules](#)